

Lab 1: Night Shift — Restore the Link (Storm Outage)

Role: You are the on-duty engineer at the Satellite Control Center (SCC).

Time: 02:14. Your phone rings. A gateway operator says only: `"GW2 link is dead. Zero."`

That is all you get. No diagnosis. A severe localized storm is sitting over VSAT7. Diagnose it yourself, stage by stage.

1. The system

- Path: GW2 (gateway) <> GEO1 (36,000 km, bent-pipe) <> VSAT7 (user terminal)
- Service contract: **≥ 35 Mbps** downlink, Ku-band, 36 MHz transponder.
- The satellite is a **transparent bent-pipe**: no onboard demod/remod, no routing. What goes up comes down, plus noise.
- Initial state: **total outage, 0 Mbps**. The link was left in a bandwidth-limited 256APSK 9/10 configuration. The storm brings heavy rain fade; wind has drifted the VSAT7 antenna.
- You do NOT have physical access. You have an API — and three stages to clear.

2. The API (your only window into the link)

Base URL: `https://sposc-lab1.onrender.com`

Each student gets an isolated simulator: always use your own ID (e.g. Neptun code, lowercase).

Use it on every call, otherwise you share the shared session with anyone who forgot their ID.

One call shows you everything. Form a hypothesis, then query:

```
curl -s "https://sposc-lab1.onrender.com/status?student=YOUR_ID" | python3 -m json.tool
```

Returns `thr_mbps` + status (down/up) + your game stage (1–4), plus full blocks:

`rf` (C/N vs lock threshold, G/T, HPA, MODCOD, Eb/N0 vs required, BER, frame loss, bandwidth),

`tcp` (RTT, window vs ready-computed BDP, SACK, PEP, ARQ),

`ttc` (orbit, station-keeping, payload health).

Filtered views of the same numbers (optional):

```
curl -s "https://sposc-lab1.onrender.com/tools/rf?student=YOUR_ID" | python3 -m json.tool
```

curl

Tip: the helper scripts do this for you — `python scripts/status.py` polls the full

status live, `python scripts/command.py cmd ... sends commands.`

Step 2 — intervene. Every command MUST carry your hypothesis in rationale

(min 15 characters), or it is rejected. It is logged and graded.

```
curl -s -X POST https://sposc-lab1.onrender.com/control \
```

-H

Reset your own session to baseline (does not affect others, needs no rationale):

```
curl -s -X POST https://sposc-lab1.onrender.com/control \
```

-H

Budget: 30 control commands per student (reset excluded, reset does NOT refill it).

The intended solve is 6 commands — full marks require ≤ 8 with a real hypothesis each. Blind scripts will burn the budget.

Allowed cmd values:

cmd	value example	what it does
`set_hpa`	`50`-`100`	HPA transmit power % (EIRP; >96 saturates)
`set_modcod`	`QPSK_1/2`, `QPSK_3/4`, `8PSK_5/6`, `16APSK_5/6`, `256APSK_9/10`	modulation + coding
`set_freq_ghz`	`12.2`, `14.0`, `22`, `60`	carrier frequency
`set_tcp_window_kb`	`64`-`8192`	TCP advertised window
`set_sack`	`1` / `0`	Selective Acknowledgments on/off
`set_pep`	`1` / `0`	Performance Enhancing Proxy on/off (spoofing + split-TCP)
`set_link_arq`	`1` / `0`	link-layer ARQ on/off
`repoint`	—	repoint ground antenna + inspect LNA (restores G/T)
`reboot`	—	reboot payload (8 s outage, your session only)
`set_payload_mode`	`BENT` / `REGEN`	payload processing mode
`reset`	—	reset YOUR session to baseline

The API replies {"ack": ..., "thr_mbps": ..., "status": "down"|"up", "stage": 1-4}.

On success it also sends a short message confirming the restore.

up (≥ 35 Mbps) means you fixed it.

Always check the student field in replies — if it is not your ID, you forgot to send it.

3. The three stages (your mission)

Stage 1 — Re-align + overcome rain (physical channel & control segment).

Heavy rain absorbs/scatters Ku-band energy; wind drifted the VSAT7 antenna (low G/T). Carrier power C fell below noise N, E_b/N_0 below threshold, modem unlocked, 0 Mbps.

- Verify via TT&C that GEO1 orbit/payload is healthy (it is — the fault is on the ground).
- repoint the antenna (restores G), then raise HPA (EIRP) into the ~88–95% linear window to burn through rain + FSPL. Max power (>96%) saturates — more is not always better.
- Gate: /tools/rf shows locked: true (C/N above threshold). Throughput is still ~0 because of Stage 2.

Stage 2 — Fall back MODCOD (PHY & link layer).

Carrier is locked but frames drop: the link is stuck in bandwidth-limited 256APSK 9/10, which needs ~18 dB E_b/N_0 you do not have under rain.

- Shift to a power-limited regime: robust QPSK. QPSK_1/2 always decodes (safe) but caps at 28 Mbps — too slow for the contract. QPSK_3/4 is the sweet spot: decodes with margin and delivers 42 Mbps.
- Higher orders (8PSK and up) stay on the BER cliff under rain.

- Gate: `/tools/rf` shows `frame_loss_pct`: 0 and clean BER. Throughput returns but stalls at a few Mbps because of Stage 3.

Stage 3 — PEP + window scaling (transport & network).

Frames are clean but TCP stalls: $\text{GEO RTT} \sim 542 \text{ ms} \times \text{rate} = \text{BDP} \approx 2.8 \text{ MB}$, while the default 64 KB window exhausts long before ACKs return; residual errors halve `cwnd` (Reno).

- Activate PEP (spoofing + split-TCP), enable SACK (isolate wireless loss from congestion), scale the window $\geq 2500 \text{ KB}$ (4096 recommended).
- Window alone caps $\sim 5.5 \text{ Mbps}$; PEP without SACK caps $\sim 12\text{--}14 \text{ Mbps}$. All three together fill the pipe.
- Gate: throughput ramps smoothly to $\geq 35 \text{ Mbps}$. API returns status: up + success message. You are done.

4. What the shift log says (unverified rumors — trust numbers, not stories)

- "Storm cell right over VSAT7, rain like crazy." (night tech — this one is true)
- "Antenna took a beating in the wind, G/T looks off." (night tech — also true)
- "Just crank HPA to 100, more power fixes everything." (previous shift — saturation says hi)
- "256APSK is the fastest, leave it." (intern — fastest way to stay at 0 Mbps)
- "QPSK 1/2 is bulletproof, good enough, right?" (intern — bulletproof but $28 < 35$)
- "Could we move to 22 GHz? Heard there's spare spectrum." (manager — absorption peak)
- "Why not handover to the next satellite like Starlink does?" (manager — GEO does not handover. Ignore.)

5. Your tasks (90 min)

1. **Baseline (15 min).** Query `/status` (note stage: 1, 0 Mbps), then `/tools/ttc` (is the satellite the problem?), `/tools/rf` (C/N vs lock? G/T? Eb/N_0 vs MODCOD requirement? BER? frame loss?), `/tools/tcp` (RTT? window vs BDP?). Record every field. Compute:

- Free-space path loss at 12.2 GHz / 35,786 km (needs `orbit_km` + `freq_ghz` from `/tools/rf`). Is path loss the problem, or rain + pointing on top of it?
- Bandwidth-Delay Product (BDP) from observed RTT and contracted rate (needs `rtt_ms` from `/tools/tcp`). Compare BDP against the 64 KB window.
- Link-budget check (needs `/tools/rf`): is C/N vs lock threshold, Eb/N_0 vs MODCOD requirement, BER, `frame_loss_pct`, G/T, `hpa_pct` healthy or not? Which stage does each number implicate?
- Classify per stage: Stage 1 power-limited (C/N)? Stage 2 bandwidth- vs power-limited (MODCOD)? Stage 3 transport-limited (BDP)?

2. **Map to layers (15 min).** For each candidate fix in the table above, state which segment (space / ground / control) and which OSI layer (PHY / link / network / transport) it acts on. Which layer do your numbers actually implicate at each stage?

3. **Intervene (45 min).** You have 30 commands and every one needs a rationale — it is logged and graded. Follow the stages: (1) repoint + HPA → RF lock; (2) MODCOD fallback → 0% frame loss; (3) PEP + SACK + window → link up. Propose a hypothesis, send ONE command, re-check status, record. Repeat. Reach up.

- Only **one combination** restores the contract. The rest do nothing or make it worse (saturation, absorption peaks, ARQ latency, outage, slow-but-capped QPSK_{1/2}). Brute-forcing without a hypothesis is rejected by the API and penalized in grading.

4. **Report (15 min, 1 page max).** Submit:

- Initial status dump + your three calculations from (1).
- Command log with hypothesis → result for each attempt, labeled by stage.

- Final status dump showing up + success message.
- Explanation: why did each stage's fix work, and why did at least three alternatives fail — using Ch.1 terms: bent-pipe vs regenerative, GEO delay, FSPL, rain fade, gaseous absorption, E_b/N_0 vs BER, power- vs bandwidth-limited, BDP / slow-start / Reno / PEP / SACK.

6. Rules

- Do not read sat_api.py. Treating it as the answer key is a fail.
- Do not assume the fault is where the rumors say it is — except the storm, which is real.
- A healthy TT&C with a dead throughput is data, not a contradiction. Think layers and stages.
- AI tools are permitted, but every command needs your own stated hypothesis — it is logged and graded. An AI that tells you *what to try next* is being used well; one you ask for the final answer will guess decoys (max HPA, 256APSK, 22 GHz).
- Use only YOUR student ID; solving in another session does not count.

Grading (10 pts)

- Correct BDP + link-budget verdict per stage (3)
- Correct layer/segment mapping (2)
- Reaching up with ≤ 8 commands and a logged hypothesis per command (3)
- Correct Ch.1 explanation of each stage fix + 3 failed alternatives (2)